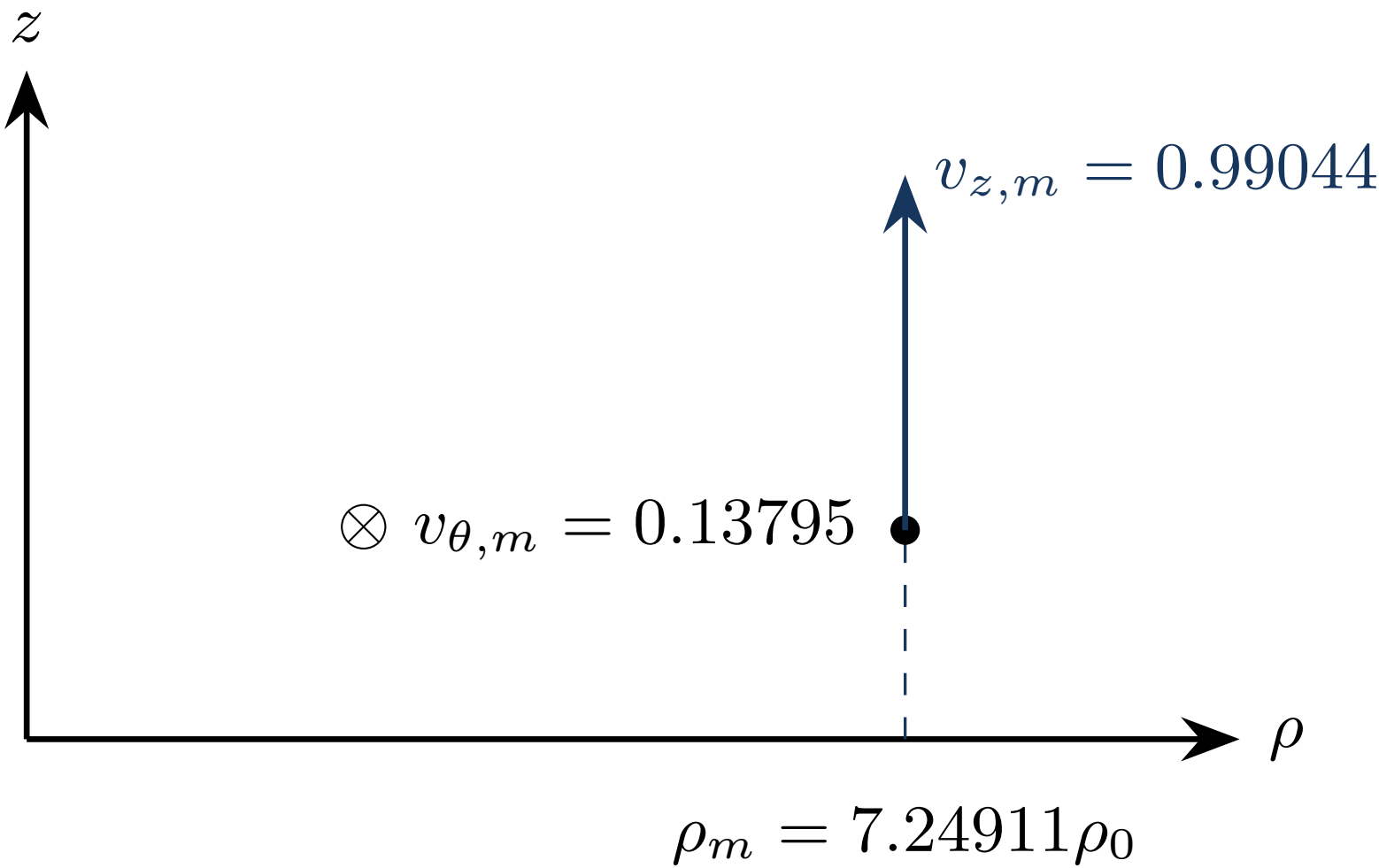外转折处：$v_\rho = 0$，总速率仍为 1